

Certain requirements (business rules) cannot be fully captured by the UML. Consider the following requirements which you must specify in the Object Constraint Language (OCL).

“A task cannot have more than 20 sub-tasks.”.

context Task

inv: self.subtask → size() <= 20

“The number of open tasks without a due date should not exceed 50.”

context TaskDB

inv: self.task → select(t_completed = false and dueDate = null) → size() <= 50

“The limit for open tasks for each collaborator category is a positive integer.” (i.e. Senior collaborators are limited to 2 open tasks.)

context Collaborator

inv: self.juniorcollaborator → forAll(j: j_max_Tasks > 0) and
self.intermediatecollaborator → forAll(i: i_max_Tasks > 0) and
self.seniorcollaborator → forAll(s: s_max_Tasks > 0)

“No collaborator must be overloaded”. (the number of assigned tasks that are open should not exceed the limit.)

context Collaborator

inv: self.juniorcollaborator → forAll(j: j_max_Tasks >= num_open_tasks) and
self.intermediatecollaborator → forAll(i: i_max_Tasks >= num_open_tasks) and
self.seniorcollaborator → forAll(s: s_max_Tasks >= num_open_tasks)